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Well Integrity: The Impact of Testing and Monitoring on CO₂ Storage Well Integrity During the Injection Phase – A Quantitative Risk Approach

P. Lucchetti, DNV, Rio de Janeiro, Rio de Janeiro, Brazil; D. Buchmiller, DNV, Oslo, Norway; B. Baltar, J. Domingues, C. Bleser Serta, G. De Abreu, and L. Oliveira, DNV, Rio de Janeiro, Rio de Janeiro, Brazil

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Carbon capture and storage (CCS) is an effective technology with the potential to significantly reduce carbon dioxide (CO₂) emissions from major human-made sources. As part of CCS projects, storage of CO₂ generally takes place in geological formations, either in saline aquifer formations or depleted oil and gas reservoirs (Daniels, et al., 2023). In the subsurface storage, there are two types of wells; firstly, injection and monitoring wells facilitating the injection and storage of CO₂ in the reservoir(s) and secondly legacy wells, which are either previously used Oil and Gas wells for exploration or production purposes, which are intended to be plugged and abandoned prior to CO₂ storage. The CO₂ injection and monitoring wells are typically constructed similarly to Oil and Gas water or gas injection wells, with the intent to safely and securely inject CO₂ according to industry regulations such as ISO 27914:2017 and DNV RP J203:2019-09. The specific well design of the injection and monitoring wells is regulated on a national level with additional operator specifications being added onto industry standards. CO₂ injection and monitoring wells differ from Oil and Gas injection wells (Todorovic, et al., 2024) and can be distinguished by the following factors:

- Increased risk of well related leakage as the project time transpires, CO₂ injection pressures increase and migration of the CO₂ plume on both injection and legacy wells.
- Corrosive environment, where fluids in the wellbore and at the near wellbore injection area influence material selection, metallurgy, elastomer, cement and other well barrier integrity and performance. Similarly, the composition of the injected CO₂ fluids may include impurities impacting the well barrier elements integrity.
- Cyclical loading on the injection wells, with potential start/stop injection operations causing pressure cycling, thermal loading including potential Joule-Thompson behavior, which could impact both the well tubular and cement integrities.
- Increased demands for CO₂ well injectivity performance relative to water and gas injection on Oil and Gas wells with completion efficiency and injection component operation.
- Increased monitoring functionality desires as measurement monitoring and verification (MMV) programs look to properly manage CO₂ injectivity and well containment.

These distinguishing factors and demands underscore the importance for well integrity management of CO₂ injection wells.

Legacy wells within the CO₂ storage complex also require containment assurance and integrity assessments (ISO 27914:2017 and DNV RP J203:2019-09). The integrity and risk assessments for legacy wells aim at capturing both the potential consequence of loss of containment through legacy wells and the probability of such incidents occurring. Gathering integrity data on legacy wells and fully mapping and assessing their integrity may be challenging as it is not uncommon for the legacy wells to be decades old, where the operators data of the wells may not exist or may not have kept records from previous well drilling, operations and plugging and abandonment operations.

Introduction

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Brown, S., et al. summarizes the industry experience with risk assessment methodologies within CO₂ storage wells, both for injection and legacy wells: "Many RA methodologies have been adapted or developed to assess leakage risk at GCS sites. These include the use of Features, Events, and Processes (FEPs) (Quintessa 2014), Failure Modes and Effect Analysis (FMEA) (Guo et al., 2018), the Certification Framework (CF) (Oldenburg et al., 2009), the Bowtie method (Bourne et al., 2014), as well as tools developed by the United States (U.S.) Department of Energy (DOE) Office of Fossil Energy and Carbon Management's National Risk Assessment Partnership (NRAP) (Carroll et al., 2020; Carroll et al., 2016)". Currently, the most prevalent of these methods is the Bowtie method, which has vast industry traction and is a preferred method among operators, (Tucker, et al., 2013). When performing Bowtie analyses or traditional containment risk assessments, the probability of failure of the individual well barrier elements and the well barrier envelopes are often assessed qualitatively using library look-up data, for example (Daniels, et al., 2023). Other quantitative methods are Monte Carlo simulations based on the qualitative library data, producing distribution curves of potential probabilities, which are used in the risk and uncertainty discussions and decision-making process. Improving upon the industry methods available and implementing a reliability-based software solution, a time dependent reliability model, with an extensive history from operational Oil and Gas wells facilitates more refined and robust well integrity leak probability risk assessments (Oliveira et al., 2022).

Measurement Monitoring and Verification Plan – CO₂ wells

Injection and monitoring wells are vital instruments in MMV plans for CO₂ storage projects, where either continuous or sporadic intermittent data is gathered to understand the reservoir response and CO₂ plume and well integrity (UK Government, 2022). Injection and monitoring wells are tailored to support robust MMV activities and CCS project developers typically evaluate the requirements for MMV and desired functionality. There is a wide range of MMV technologies, for example downhole gauges and temperature and pressure sensors and wellbore and near wellbore sensors (DNV KEM Programme, 2024), where the CCS wellbore monitoring plan aims to:

- Validate reservoir simulation models on reservoir response and dispersion of the CO₂ plume.
- Assess injection performance and near wellbore.
- Map downhole fluid characteristics and chemistry.
- Ensure well integrity by tracking pressure and temperature for signs of leaks, blockages, or mechanical damage.
- Monitor injection rates and pressures to verify they remain within operational limits.
- Assess geomechanical responses by observing subsurface pressure, temperature, and rock deformation.
- Detect induced seismicity through seismic monitoring near the wellbore.
- Identify and respond to abnormal or emergency events, such as sudden pressure increases, to protect personnel and the environment.
- Verify long-term CO₂ containment by monitoring subsurface conditions to ensure secure storage.

From a well integrity perspective, the monitoring data, in addition to the well barrier testing constitute key activities performed by CO₂ storage operators.

Well Integrity Management Principles

Both ISO 16530-1:2017 and NORSOK D-010:2021 emphasize the importance of risk-based well integrity management throughout the entire well life cycle. ISO 16530-1:2017 highlights that the well operator should

assess failure risks and define monitoring and surveillance requirements to ensure that wells operate within safe limits. These requirements should be documented in the Well Integrity Management System (WIMS), and based on risk-based considerations, such as reliability assessments and mean-time-to-failure analysis. Meanwhile, [NORSOK D-010:2021](#) stipulates that in the event of a well barrier failure or loss, immediate measures must be taken to prevent escalation by activating the secondary barrier.

Given the challenges posed by CO₂ injection operations, it is essential to adopt quantitative risk analysis models rather than relying solely on qualitative methods, which often involve considerable uncertainty. Unlike traditional approaches, quantitative methodologies enable a more accurate evaluation of well risks, offering a real-time snapshot of the well's status and estimating the remaining time before risk levels exceed tolerable limits. By incorporating barrier component failure data, testing and monitoring conditions, and the current status of barrier elements, this approach provides a more robust foundation for decision-making. It also facilitates the evaluation of monitoring's impact on risk and the identification of grace periods that may guide the scheduling of interventions, enhancing the safety and reliability of carbon storage wells.

Methodology

This study adopts the same time-dependent quantitative risk and reliability model previously developed and applied to production wells ([Oliveira, et al., 2022](#)), now adapted to the context of CO₂ injection and storage wells during the injection phase. The model is designed to support operational decision-making by quantifying the instantaneous frequency of loss of well integrity (e.g., blowout or leakage) based on the current condition and historical performance of well barrier elements. The proposed methodology establishes a robust framework for evaluating the frequency-probability of the unwanted event, recognizing that an uncontrolled release of CO₂ carries critical consequences. Accordingly, the terms ‘frequency’ and ‘risk’ are used interchangeably throughout the analysis.

The methodology consists of two main computational components:

1. Logical Modeling of Leak Paths

All potential leak paths are identified through a graph approach, where each path corresponds to a minimal cut set of barrier element failures modes that could lead to an uncontrolled release of CO₂. The well barrier schematics are used as the basis for constructing the model, following the principles outlined in [NORSOK D-010:2021](#) and [ISO 16530-1:2017](#).

2. Time-Dependent Reliability Modeling

For each failure mode, the model calculates:

- Availability and unavailability at time $t = t$
- Failure rate and failure intensity at time $t = t$

These are computed using known failure rate distributions (exponential or Weibull), and updated dynamically as new events (failures, repairs, replacements) are detected. The model distinguishes between:

- Monitored components (Type M), for which information of barrier status is always known
- Periodically tested components (Type T), for which information of barrier status is known when a periodic test is performed
- Unmonitored and untested components (Type N), for which the information of barrier status is unknown

To support decision-making, the model uses risk control limits to define acceptable, tolerable, and unacceptable risk regions. These limits are based on industry standards such as [IEC 61508:2010](#), which defines Safety Integrity Levels (SIL), and [NOROG:OLF-070:2018](#), which provides recommended practices

for the Norwegian petroleum industry. Based on these references, the Lower Control Limit (LCL) is set at 1.0×10^{-7} failures per hour (corresponding to SIL 3), and the Upper Control Limit (UCL) at 1.0×10^{-6} failures per hour (SIL 2). These values are defined under continuous demand mode, since the well is constantly required to perform its safety function of containing the injected fluid. Notably, SIL 3 is the highest level typically applied in the process industry, representing a very high degree of risk reduction.

The methodology uses three quantitative indicators to support operational decisions:

1. Instantaneous leak frequency: the current system failure rate, calculated from all possible leak paths.
2. Cumulative probability of loss of containment: the total risk of failure over the remaining life of the well.
3. Incremental Cumulative Risk (ICR): the additional risk accumulated after a detected failure, compared to normal operation without failure.

These indicators are used to define operational grace periods, which estimate the number of days remaining before the risk crosses predefined thresholds: the Lower Control Limit (LCL), the Upper Control Limit (UCL), or the ICR limit. These grace periods help determine whether planned mitigation actions are sufficient or if immediate intervention is required to restore well integrity and maintain safe operation.

The ICR is an indicator applied exclusively in scenarios where a well has already experienced a failure, while the LCL and UCL can be applied to all scenarios. It is important to note that the cumulative probability of loss of containment is calculated based on the well's end-of-life (or end-of-analysis) date; any change to this date will directly affect the results. For the full mathematical formulation and implementation details, the reader is referred to the original work (Oliveira et al., 2022), which provides the complete derivation of the reliability equations and modeling assumptions.

Case Study

Two base case studies were conducted for CO₂ wells: one considering an injection well, and the other a legacy well. The well schematics used in each case are presented in Figures 1 and 2 below, representing the typical configuration and barrier elements relevant to each well type.

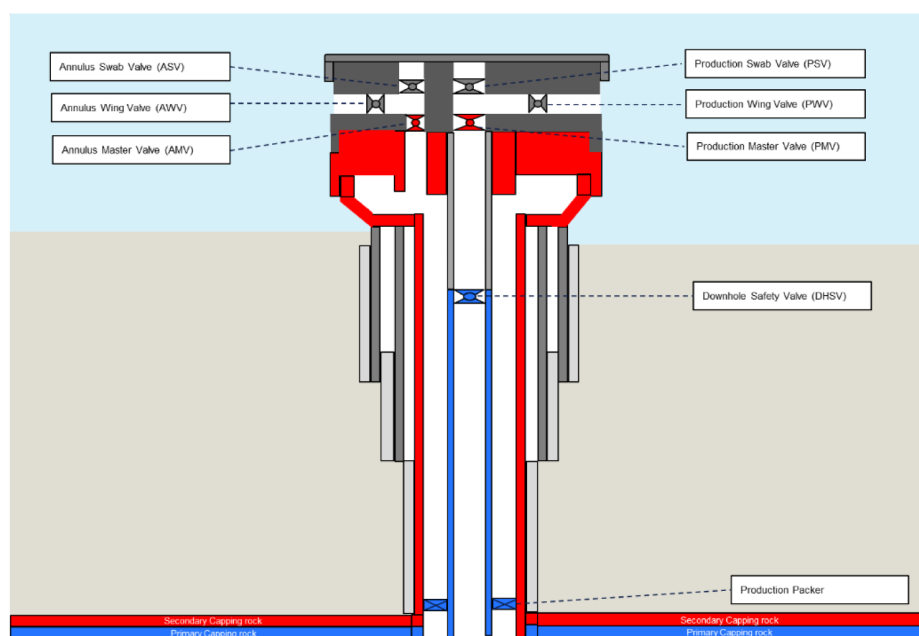


Figure 1—Schematic of the injection well (Base Case 1).

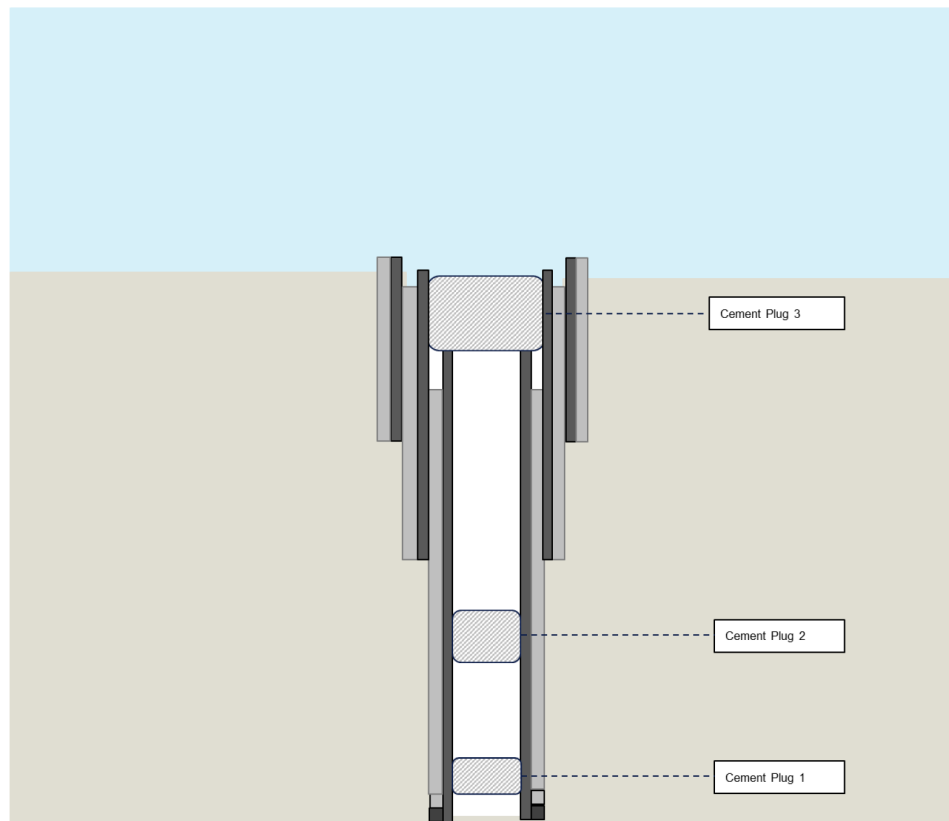


Figure 2—Schematic of the legacy well (Base Case 2).

Table 1 summarizes the key assumptions considered for both base cases.

Table 1—Assumptions used for Base Case 1 and Base Case 2 modeling.

Parameter	Injection Well	Legacy Well
Start Date	01/01/2005	01/01/1995
End Date	01/01/2055	01/01/2055
Lower Control Limit	1.00E-07	1.00E-07
Upper Control Limit	1.00E-06	1.00E-06
ICR Limit	5.26E-3	5.26E-3
Failed Components	None	None
Monitoring	None	None

The sensitivity analyses performed for each base case are presented in Table 2. These scenarios were selected to reflect real-world operational challenges and support risk-informed decision-making, each exploring a specific aspect of well integrity management. Cases involving monitoring and testing (e.g., Annulus A and Cement Plug 3) simulate strategies that operators may adopt to enhance early failure detection. Scenarios with increased failure rates represent degradation mechanisms influenced by CO₂ exposure, such as corrosion or material fatigue, helping assess the impact of conservative assumptions. Failure insertion cases (e.g., DHSV and Cement Plug 1) illustrate the consequences of unaddressed failures over time, while the repositioning of the DHSV evaluates design alternatives to reduce exposure and associated risks. Together, these cases provide insights into how operational decisions, monitoring strategies, and design choices influence well integrity risk.

Table 2—Sensitivity analysis scenarios considered for each base case.

Case Number	Sensitivities for Injection Well	Sensitivities for Legacy Well
#2	Annulus A monitored	Cement plug 3 monitored
#3	Annulus A Tested once every two years	Cement plug 3 Tested once every two years
#4	Increase 5 times on the failure rates of some primary barriers: 1. Production Packer, 2. Nipple 3. Production Casing Below Packer, 4. Tubing Anchor Assembly, 5. Tubing Below DHSV 6. Tubing Seal Receptacle	Increase in 5 times on the failure rates of the Cement Plug 1
#5	DHSV Failed on 2010/01/01	Cement Plug 1 Failed on 2020/01/01
#6	Change in the position of DHSV (from 50 to 450m below)	-

In addition to the assumptions presented in [Table 1](#), the following were also considered:

General Assumptions.

1. Due to the lack of failure data for CO₂ injection wells, the failure rates assigned to each failure mode were based on historical data from oil and gas wells;
2. The cement was considered to act as a barrier only within the first 60 meters of the well, as per general Oil and Gas Industry practices;
3. The consideration of the intermediate casing (intermediate casing that confines Wellbore Annulus B) as a barrier element depends on its ability to withstand injection fluid pressures in accidental scenarios involving failure of the production casing (communication between Annulus A and B). Since there is no information confirming its resistance to the maximum injection pressure, as a conservative approach, the intermediate casing was not classified as a safety barrier element;
4. The analysis considers five key failure modes relevant to well integrity and system performance. These failure modes, along with their abbreviations, are: FTC (Fail to Close); LCP (Leak in Closed Position); INL (Internal Leakage); EXL (External Leakage); ANS (Anchor Snagging); It is important to note that not all elements exhibit all the identified failure modes. The applicability of each failure mode depends on the specific function and configuration of the component being evaluated.

Assumptions Considered for the Injection Well.

1. For sensitivity cases involving the monitoring and testing of Annulus A, all barrier elements connected to Annulus A (primary completion) were considered as monitored or tested. Additionally, the barrier elements linking Annulus A to the reservoir were included under the same assumption. This approach is based on the premise that any leakage to the external environment would be detectable due to the pressure differential.

Results and Discussions

Base Case 1: Injection Well

The graph-based analysis of the well configuration identified more than 1000 minimal cut sets, which were grouped into 58 distinct leak paths. However, a significant portion of the overall risk is concentrated on just a few of these leak paths, which are presented in [Table 3](#). The top five leak paths account for over 95% of the total frequency of uncontrolled CO₂ release.

Table 3—Most Important Leak Paths for the Injection Well (Base Case 1)

#	Leak Path	Frequency	Contribution
#1		1.78E-07	87.2%
#2		4.88E-09	2.4%
#3		4.76E-09	2.3%
#4		3.34E-09	1.6%
#5		3.21E-09	1.6%

The most important leak path is of second order, (i.e., two barrier layers), with 12 minimal cut sets, and has a contribution of 87.2% of the total risk of CO₂ release. The second and third most significant paths (LP#2 and LP#3) are second and fourth order and contribute less, 2.4% and 2.3% of the total risk, respectively, comprising 2 and 40 minimal cut sets. Both paths #4 and #5 contribute to 1.6%, and are fourth and third-order, with 60 and 16 minimal cut sets each. It can be noted that the primary barrier elements (Injection Packer, Nipple, DHSV, Tubing Anchor Assembly, Tubing Below DHSV, and Tubing Seal Receptacle) appear in two of the top five leak paths, presenting themselves as important candidates for well integrity focus, as well as the other barrier elements presented in the most important leak path.

The results of uncontrolled leak frequency and cumulative risk probability over the evaluation interval are presented in Figure 3. As of today, the Risk Status is $2.0 \times 10^{-7}/h$, representing the likelihood of a loss of containment in the well at the current moment. The cumulative probability of uncontrolled leak throughout the interval (from today until the well's end of life on January 1st, 2035) is 2.24×10^{-2} , indicating the overall probability of a loss of containment during the remaining operational life of the well.

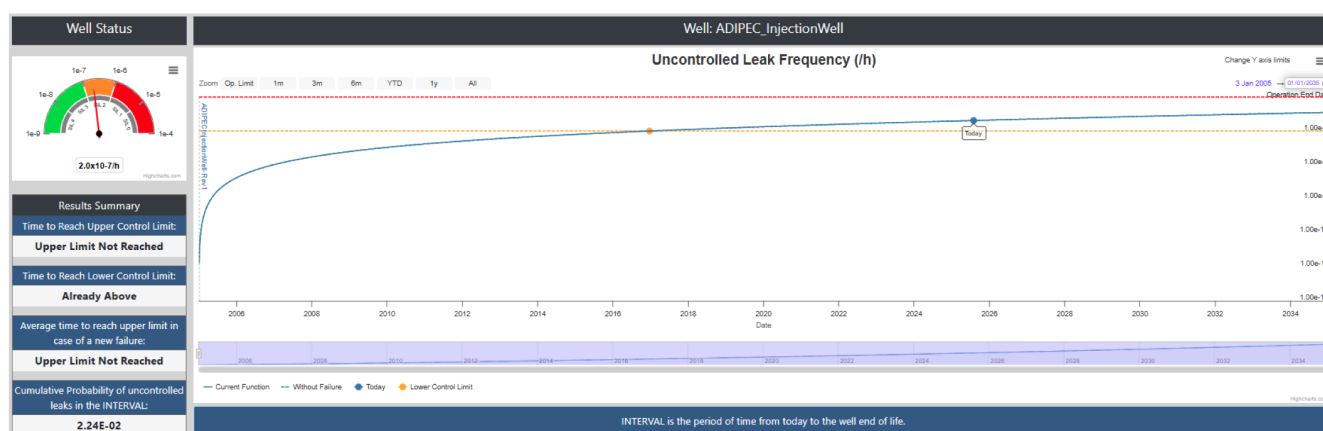


Figure 3—Uncontrolled Leak Frequency for the Injection Well (Base Case 1).

It is also observed that, under the Base Case scenario, the well's leak frequency lies above the Lower Control Limit (yellow curve = 1×10^{-7}) but below the Upper Control Limit (red curve = 1×10^{-6}).

It is important to highlight that the cumulative probability is dependent on the defined end-of-life date of the well. If the operational period was extended beyond 2035/01/01, the leak frequency could potentially reach or exceed the upper control threshold.

Table 4 presents the top 10 failure modes that have had the greatest impact on the well's integrity, ranked by significance. These failure modes are presented within the two primary leak paths, underscoring the most critical failure modes and associated barrier elements that contribute to the loss of well integrity and that demand particular attention due to their relevance and potential consequences.

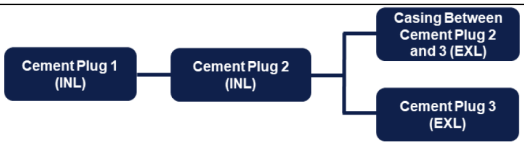
Table 4—Most Important Contributor Failure Modes for Well Integrity (Base Case 1)

#	Barrier Element	Failure Mode
#1	Tubing Below DHSV	Internal Leakage
#2	Casing A	External Leakage
#3	Wellhead Packoff	External Leakage
#4	XT Connector	External Leakage
#5	Casing Hanger	External Leakage
#6	Tubing Seal Receptacle	Internal Leakage
#7	Injection Packer	Internal Leakage
#8	DHSV (Downhole Safety Valve)	Leak in closed position
#9	DHSV (Downhole Safety Valve)	Fail to close
#10	PMV (Production Master Valve)	Leak in closed position

Base Case 2: Legacy Well

In the case of the legacy well (Base Case 2), a total of seven minimal cut sets were identified, which were grouped into three distinct leak paths. These leak paths represent the most critical contributors to the overall risk of uncontrolled release. As shown in Table 5, the most significant leak path is through Cement Plug 1 to the casing between Cement Plug 1 and 2, formed by 1 minimum cut set of second order, with a frequency of 1.47×10^{-8} and a contribution of 75.5% to the total risk. The second most important path involves a sequential failure from Cement Plug 1 to Cement Plug 2 and then to Cement Plug 3 or Casing between Cement Plugs 2 and 3, formed by 2 minimum cut sets of third order, with a frequency of 4.78×10^{-9} , contributing 24.5%. The third path, involving the primary and secondary caprock, formed by 4 minimum cut sets of second order, and has a negligible frequency of 2.59×10^{-12} , contributing only 0.01% to the overall risk. These results highlight the dominant role of cement plug integrity in the current case.

Table 5—Most Important Leak Paths for the Legacy Well (Base Case 2)

#	Leak Path	Frequency	Contribution
#1		1.47E-08	75.5%
#2		4.78E-09	24.5%
#3		2.59E-12	0.01%

The results of uncontrolled leak frequency and cumulative risk probability over the evaluation interval for the legacy well are presented in Figure 4. The current Risk Status is $1.9 \times 10^{-8}/h$ and the cumulative probability of uncontrolled leak throughout the interval (from today until the well's end of life on January 1st, 2035) is 2.04×10^{-3} , indicating the overall probability of loss of containment during the remaining operational life.

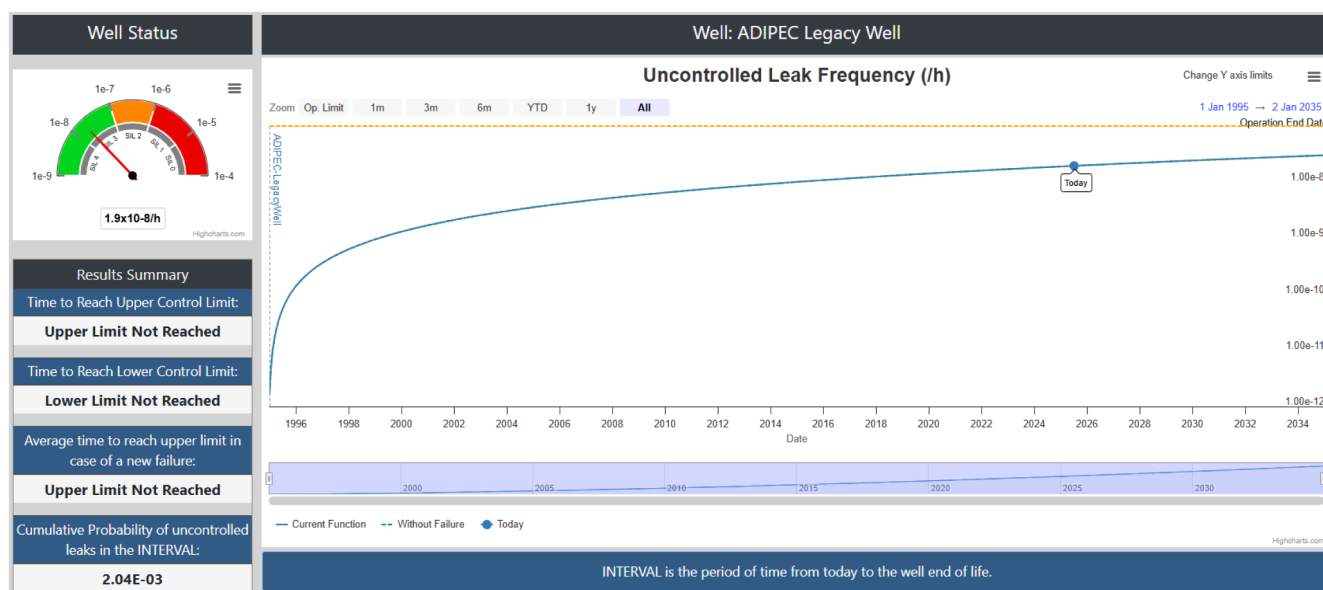


Figure 4—Uncontrolled Leak Frequency for the Legacy Well (Base Case 2).

Additionally, the well's leak frequency lies below the Lower Control Limit (yellow curve = 1×10^{-7}). Neither the lower nor the upper control limits are reached throughout the considered timeframe up to the well's end of life.

Table 6 highlights the most critical failure modes contributing to the loss of well integrity in the legacy well scenario, with each failure mode directly associated with one of the three main leak paths identified in Table 5. The dominant contributor is internal leakage in Cement Plug 1, which initiates both the first and second leak paths. The casing between Cement Plug 1 and Cement Plug 2 is the second most significant contributor, as part of the most important leak path. This is followed by Cement Plugs 2 and 3, and the casing between Cement Plug 2 and Cement Plug 3, which are part of the second most important leak path. Primary caprock and cement, along with secondary caprock and cement, are associated with the third leak path, which has the lowest contribution. These results emphasize that Cement Plug 1 is the most critical barrier element in this case.

Table 6—Most Important Contributor Failure Modes for Well Integrity (Base Case 2)

#	Barrier Element	Failure Mode
#1	Cement Plug 1	Internal Leakage
#2	Casing Between Cement Plug 1 and Cement Plug 2	External Leakage
#3	Cement Plug 2	Internal Leakage
#4	Cement Plug 3	External Leakage
#5	Casing Between Cement Plug 2 and Cement Plug 3	External Leakage
#6	Primary Caprock	Internal Leakage
#7	Primary Cement	Internal Leakage
#8	Secondary Caprock	External Leakage
#9	Secondary Cement	External Leakage

Sensitivities Results

The sensitivity results for the injection well are presented in Table 6a. Based on the results, it is evident that different sensitivity scenarios present various frequencies and cumulative probabilities of uncontrolled leaks in the injection well, with each scenario having a distinct impact on overall well risk. The base case shows

a leak frequency of $2.04\text{E-}07$ and a cumulative probability of $2.24\text{E-}02$. Scenarios involving monitoring or testing of Annulus A (cases #2 and #3) lead to substantial risk reductions, with cumulative probabilities decreasing by up to 87.0 and 84.5% compared to the base case. Conversely, increasing the failure rates of key primary barriers (case #4), which could be influenced by the presence of CO_2 and water, results in a significant risk escalation, with cumulative probabilities rising by as much as 159.0%.

Table 6a—Summary of Sensitivity Results for the Injection Well

Case	Description	Uncontrolled Leak Frequency Today	Time to reach Lower Control Limit (days)	Time to reach Upper Control Limit (days)	Time to Reach ICR Limit from Today (days)	Cumulative Prob. Of Uncontrolled Leak	Cumulative Prob. % versus Base Case
#1	Base Case	$2.04\text{E-}07$	Already above	UCL Not Reached	-	$2.24\text{E-}02$	Base Case
#2	Annulus A monitored	$1.70\text{E-}08$	LCL Not Reached	UCL Not Reached	-	$2.91\text{E-}03$	-87.0%
#3	Annulus A Tested	$2.43\text{E-}08$	LCL Not Reached	UCL Not Reached	-	$3.47\text{E-}03$	-84.5%
#4	Increase 5 times on the failure rates of some primary barriers: 1. Production Packer, 2. Nipple 3. Production Casing Below Packer, 4. Tubing Anchor Assembly, 5. Tubing Below DHSV 6. Tubing Seal Receptacle	$6.50\text{E-}07$	Already above	UCL Not Reached	-	$5.80\text{E-}02$	159.0%
#5	DHSV failed on 2010/01/01	$2.41\text{E-}07$	Already above	UCL Not Reached	7788	$2.55\text{E-}02$	13.7%
#6	Change in the position of DHSV (from 50 to 450m below)	$1.86\text{E-}07$	Already above	UCL Not Reached	-	$2.14\text{E-}02$	-4.6%

The DHSV failure on 2010/01/01 (case #5) leads to a 13.7% increase in cumulative probability. In failure scenarios, the risk indicator used is the ICR. For the DHSV failure case, the ICR results in a grace period of 7788 days, meaning that this is the maximum allowable time to repair the failure before the risk reaches unacceptable levels, assuming no additional failures occur during this period.

A sensitivity analysis was also conducted by changing the DHSV installation depth. When the DHSV was repositioned 450 meters deeper, a 4.6% causes a reduction in the cumulative leak probability, since lowering the DHSV, reduces the length of tubing located below it. As the tubing failure rate is defined per meter, a shorter section means fewer opportunities for failure. Additionally, for a leak to occur through the tubing above the DHSV, the valve must first fail to close, acting as an additional barrier element. Therefore, this configuration changes directly the leak frequency by reducing the exposure length of tubing below the DHSV.

Since the cumulative probability is highly dependent on the defined end-of-life date of the well, all sensitivity cases considered a fixed end date of 2035/01/01. Any change to this date will directly affect

the results. These findings highlight the critical importance of barrier integrity and reinforce the value of quantitative risk assessment in evaluating well integrity risks.

The sensitivity results for the legacy well are presented in Table 7. In the case of the legacy well, the sensitivity analysis also reveals notable variations in risk depending on the scenario evaluated. The base case presents an uncontrolled leak frequency of $1.9 \times 10^{-8}/\text{h}$ and a cumulative probability of 2.04×10^{-3} . Some improvements are observed when Cement Plug 3 is either monitored or tested (cases #2 and #3), resulting in risk reductions of approximately 12%.

Table 7—Summary of Sensitivity Results for the Legacy Well

Case	Description	Uncontrolled Leak Frequency Today	Time to reach Lower Control Limit (days)	Time to reach Upper Control Limit (days)	Time to Reach ICR Limit from Today (days)	Cumulative Prob. Of Uncontrolled Leak	Cumulative Prob. % versus Base Case
#1	Base Case	1.95E-08	LCL Not Reached	UCL Not Reached	-	2.04E-03	Base Case
#2	Cement plug 3 monitored	1.73E-08	LCL Not Reached	UCL Not Reached	-	1.78E-03	-12.7%
#3	Cement plug 3 Tested	1.74E-08	LCL Not Reached	UCL Not Reached	-	1.79E-03	-12.2%
#4	Increase in 5 times on the failure rates of the Cement Plug 1	1.03E-07	Already above	UCL Not Reached	-	1.01E-02	397.9%
#5	Cement Plug 1 Failed on 2020/01/01	1.67E-07	Already above	UCL Not Reached	220	5.12E-03	151.6%

Additionally, the integrity of Cement Plug 1 proves to be critical. When its failure rate is increased five times, the cumulative probability rises sharply by nearly 397.9%, indicating a substantial impact on well integrity. Similarly, a failure of Cement Plug 1 on 2020/01/01 leads to a 151.6% increase in cumulative probability, with the risk already exceeding the lower control limit but remaining below the upper control limit. Once the failure is included, the ICR indicator should be evaluated, and it presents a grace period of 220 days until the well reaches an unacceptable risk level. These results emphasize the influence that certain barriers, particularly Cement Plug 1, can have on overall well integrity.

The same consideration applied to the injection well is also valid for the legacy well. Since the cumulative probability is highly dependent on the defined end-of-life date of the well, all sensitivity cases assumed a fixed end date of 2035/01/01. Any change to this date will directly affect the results.

The results from both the injection and legacy wells clearly demonstrate the value of monitoring and testing strategies in managing well integrity. Sensitivity analyses revealed that even the monitoring of a single barrier, such as Cement Plug 3, can lead to meaningful reductions in cumulative risk. Additionally, failures or degradation in critical components, like DHSV or Cement Plug 1, can significantly increase the probability of uncontrolled leaks.

By quantifying the time remaining before risk thresholds are exceeded, the model provides actionable insights for planning interventions. This enables operators to prioritize repairs and allocate resources based on the relative impact of each failure scenario, considering the role of each barrier within the leak paths.

Beyond safety improvements, this approach also contributes to cost efficiency. By identifying which failures pose the greatest risk, operators can avoid unnecessary interventions and focus efforts where they are most impactful, optimizing workover campaigns, reducing downtime, and improving rig logistics. These benefits reinforce the value of integrating quantitative risk analysis into well integrity management.

Conclusions

This study demonstrates the effectiveness of a quantitative, time-dependent risk assessment methodology in evaluating the integrity of CO₂ injection and legacy wells and clearly shows the critical role of monitoring and testing strategies in managing well integrity. By applying a reliability-based model to both well types, it was possible to identify the most critical leak paths and barrier elements, quantify the impact of monitoring and testing strategies, and define operational grace periods that support informed decision-making.

The use of grace periods, such as the time to reach the upper control limit or the ICR threshold in failure scenarios, provides a practical and time-bound framework for decision-making while prioritizing safety. By estimating the number of days remaining before risk thresholds are exceeded, operators can prioritize the most critical failures across their well portfolio. Each failure impacts well integrity differently, as each barrier element is part of one or more leak paths with varying levels of importance.

Importantly, this approach enables operators to assess and compare risks across their entire well portfolio. By identifying which failures have the greatest impact on integrity, operators can better plan and prioritize workover campaigns, optimize rig logistics, and allocate resources more efficiently. This not only enhances operational safety but also delivers financial benefits by reducing the number of interventions and focusing efforts where they are most needed. As the industry moves forward with digitalization and refined risk methods, this strategy supports a more proactive and data-driven approach to well integrity management.

Ultimately, the findings reinforce the importance of adopting quantitative risk analysis as a standard practice in CO₂ storage projects. It provides a more objective, data-driven foundation for well integrity management and supports the industry's transition toward digital, risk-informed decision-making.

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